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Work Experience

Software Development Engineer Intern *Sept 2026 - Dec 2026*
AWS (Amazon Web Services)

- Incoming SDE Intern on Amazon SageMaker AI, working on distributed training and post-training systems for large-scale foundation models across NVIDIA GPU and AWS Trainium environments.

Software Engineering Intern *Jun 2026 - Aug 2026*
AT&T

- Engineered a scalable backend anomaly detection service using **asynchronous FastAPI endpoints**, integrating an **Extended Isolation Forest (EIF)** model to automatically detect and suspend telecom fraud.
- Built automated **Azure ETL pipelines** to ingest **12M+ phone records** from Snowflake to PostgreSQL, executing bi-hourly incremental upserts of user telemetry metrics to trigger batch fraud inference.
- Deployed a continuous production **MLOps pipeline** retraining models on **5M+** monthly records, achieving high-precision anomaly scoring with an estimated **<0.5% false-positive rate** to safely trigger automated suspensions.
- **Streamlined fraud operations** by implementing a dashboard, enabling cross-functional teams to monitor automated enforcement actions, view aggregated statistics about banned users, and maintain platform trust.

Machine Learning Engineer *Sep 2025 - Jan 2026*
Mercor

- Compiled external ML competitions into challenging tasks reflecting real-world ML responsibilities. Drafted detailed natural-language execution plans and implemented them in Python within a Docker environment, validating implementations against original plans and marking discrepancies. These tasks are used by researchers to train models on high-quality data.

Projects

Network Degradation Prediction Using Real Internet Telemetry *2025 – 2026 School year*

- Engineered a predictive DNS telemetry pipeline with XGBoost on **327MB** of real RIPE Atlas telemetry (Root DNS anchor) to predict network degradation **5 minutes** ahead, achieving **91% accuracy** and **40% recall**.
- Deployed on **AWS EC2** with Docker and **CI/CD** via GitHub Actions; decoupled inference from serving via cron-based background refresh, achieving **sub-100ms** load time with zero model cold-start latency.

Wildfire Spread Forecaster **Hosted on Proxmox Home-Cloud** *2025 – 2026 School year*

- Architected a 3-node bare-metal **Proxmox GPU cluster** orchestrated via **Nomad**, saving **~$400/mo** in cloud infrastructure costs and serving live microservices (dhruvq.com, fire.dhruvq.com) with zero open ports via Cloudflare Tunnels.
- Optimized geospatial pipeline throughput by implementing concurrent spatial data pre-fetching via Python **asyncio.gather** and in-memory caching, eliminating sequential API bottlenecks across USGS and NOAA endpoints.
- Engineered an async **FastAPI backend** on the cluster fusing live NASA, NOAA, and USGS geospatial data, computing a 6-hour cellular automata fire spread forecast in **<10 seconds** at **75.4% IoU accuracy**.

Apollo: Autonomous AI Research Analyst *2025 – 2026 School year*

- Built an autonomous Python pipeline to ingest arXiv cs.AI papers biweekly, applying a **three-layer filtering system** to surface the **top 25 most impactful** from ~1500 submissions per cycle, auto-deployed as GitHub Pages digests.
- Developed a low-latency RAG Telegram bot leveraging a **hybrid SQLite vector and keyword store** backed by **Gemini 2.5**, for real-time grounded querying of the curated papers in **<5 seconds**.

Education

University of Southern California *Aug 2025 - Dec 2027*
Master of Science in Electrical Engineering (Data Science and Machine Learning)

- Relevant Coursework: Probability for ECE, Computational Deep Learning, ML: Supervised Methods, Database Systems

University of California, Riverside *Sept 2022 - Jun 2025*
Bachelor of Science in Electrical Engineering

- GPA:3.77/4.0; Honors:Graduated **Cum Laude**; **Chancellor's Honors List** (all years); **Dean's Honors List** (all quarters)
- Relevant Coursework: Machine Learning & Data Mining, Digital Signal Processing, Probability & Random Variables.

Skills

Languages & Frameworks: Python, SQL, TypeScript, FastAPI, Next.js, Node.js, Snowflake, PostgreSQL

ML, Infra & other: PyTorch, XGBoost, MLOps, Azure, AWS, Docker, Proxmox, Tailscale, Cloudflare, GitHub Actions, Linux, Git & Github for version control